

Coordinate-Free Physics

Paul Gustafson

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1 Superposition

Exercise 1.1 (The monochord). ***Experiment.** A string fixed at both ends rings only at frequencies*

$$f_n = n f_1, \quad n = 1, 2, \dots,$$

and an arbitrary small displacement separates into the corresponding normal modes. Let V be the real vector space of displacements, let $K : V \rightarrow V^$ encode the restoring force, and let $M : V \rightarrow V^*$ encode kinetic energy. Assume that M is positive and that K is symmetric with respect to the natural pairing.*

Explain the observed decomposition without choosing coordinates. Show that a mode of angular frequency ω is a nonzero solution of

$$Kv = \omega^2 Mv.$$

Use the M -inner product to explain why modes with distinct frequencies are orthogonal. For an ideal string, identify the boundary condition that gives $\omega_n = n\omega_1$, and state which assumptions fail when the amplitude is large enough that superposition is no longer observed.

Exercise 1.2 (Young's fringes). ***Experiment.** Young's two-slit experiment produces alternating bright and dark bands [1]. Blocking either slit removes the bands. At a detector point, let the two alternatives contribute $a, b \in \mathbb{C}$, while the measured intensity is proportional to $|a + b|^2$.*

Explain why adding measured intensities cannot produce the dark bands. Expand $|a + b|^2$, isolate the interference term, and show that it depends on the relative phase but not on a simultaneous multiplication $(a, b) \mapsto (e^{i\theta}a, e^{i\theta}b)$. Reformulate the physical state as a ray in a complex vector space. Explain why the quotient by nonzero scalar multiplication records the experimental invariance more faithfully than any choice of phase convention.

Exercise 1.3 (One photon, two paths). ***Experiment.** In a balanced Mach–Zehnder interferometer, single photons arrive one at a time, yet one output can be dark after many trials. Inserting a phase ϕ in one arm changes the output probabilities to*

$$p_0 = \cos^2(\phi/2), \quad p_1 = \sin^2(\phi/2).$$

Let H be the two-dimensional path space and let a lossless beam splitter be represented by a unitary map $B : H \rightarrow H$.

Find a coordinate-free expression for the phase shifter using the two path subspaces and their orthogonal projections. Compose two beam splitters with this map and recover the observed probabilities. Explain why a classical mixture of upper path and lower path cannot make one output dark, and why the experiment nevertheless does not determine a preferred basis for H .

Exercise 1.4 (Stern–Gerlach splitting). ***Experiment.** A beam of silver atoms passing through an inhomogeneous magnetic field splits into two beams, not a continuous smear [2]. Repeating the experiment with a rotated magnet changes the relative counts. Model a measurement direction $n \in \mathbb{R}^3$ by a self-adjoint operator S_n on a two-dimensional complex Hilbert space.*

Find structural assumptions on $n \mapsto S_n$ which force $S_n^2 = (\hbar^2/4)\text{id}$. Deduce the two observed values $\pm\hbar/2$. If P_n^+ and P_m^+ are the positive-beam projections, show that the probability of passing both filters is $\text{Tr}(P_n^+ P_m^+) = (1 + n \cdot m)/2$. Explain the disappearance and return of a blocked beam when an intermediate, differently oriented analyzer is inserted.

Exercise 1.5 (Tomography of an open evolution). ***Experiment.** Quantum process tomography reconstructs laboratory evolutions which are not unitary: attenuation, dephasing, and relaxation all occur [3]. Let $\Phi : \text{End}(H) \rightarrow \text{End}(K)$ be the reconstructed linear map. Experiments require density operators to remain positive even when the system is entangled with an untouched auxiliary system A .*

Show that the requirement is positivity of $\Phi \otimes \text{id}_{\text{End}(A)}$ for every finite-dimensional A . Prove that it suffices to test $A \cong H$. From the positive operator obtained by applying $\Phi \otimes \text{id}$ to a maximally entangled state, derive operators $V_i : H \rightarrow K$ satisfying

$$\Phi(\rho) = \sum_i V_i \rho V_i^*.$$

Explain why preservation of total detection probability is equivalent to $\sum_i V_i^ V_i = \text{id}_H$.*

Exercise 1.6 (Chladni figures). ***Experiment.** Sand on a driven metal plate collects along sharp nodal curves, and the pattern changes abruptly as the driving frequency is varied. Let V be the displacement space and let the plate’s elastic and inertial forms be K and M .*

Explain why persistent patterns occur near the spectrum of the generalized operator $K - \omega^2 M$. Show that the sand records the zero set of an eigenvector rather than its coordinates. Explain why a geometrically symmetric plate can have degenerate modes, why a small clamp or added mass splits them, and why an arbitrary linear combination is observed when the driving frequency meets an unsplit eigenspace.

Exercise 1.7 (Malus’s law). ***Experiment.** Light transmitted through two ideal linear polarizers has intensity $I = I_0 \cos^2 \theta$, where θ is their relative angle. Represent a pure polarization by a ray in a two-dimensional complex inner product space and a polarizer by an orthogonal projection P of rank one.*

Explain the observed law by showing that the transmitted fraction from ray $[v]$ is $\|Pv\|^2/\|v\|^2$. Prove that this number depends only on the two rays and not on phases or chosen bases.

Explain why inserting a third polarizer between crossed polarizers can restore nonzero transmission.

Exercise 1.8 (Ramsey fringes). **Experiment.** *Two separated resonant pulses applied to an atom produce an oscillating excitation probability as the pulse separation or oscillator frequency is varied. Let the two energy eigenspaces be orthogonal lines in H ; model each pulse by a unitary mixing them and free evolution by the spectral projections of the Hamiltonian.*

Compose the three maps and recover a sinusoidal detection probability. Explain why the phase accumulated during the unobserved interval affects the final population. Identify the invariant relative phase and explain how the fringe period measures the energy splitting without selecting coordinates on H .

Exercise 1.9 (Neutron diffraction). **Experiment.** *A monochromatic neutron beam scattered by a crystal gives intense peaks only at particular angles. Associate an amplitude to scattering from each lattice site and let translations act on the wave's one-dimensional phase space.*

Explain why the total amplitude is a sum of translation characters. Show that constructive interference occurs precisely when the momentum transfer defines the trivial character on the crystal lattice. Recover Bragg's law after choosing a family of lattice planes. Explain why the reciprocal lattice, defined as a character group, is independent of any chosen primitive cell.

Exercise 1.10 (Interference of large molecules). **Experiment.** *Molecular beams passing through a diffraction grating produce interference fringes even when the molecules have many internal degrees of freedom. Associate a complex amplitude to each center-of-mass path and an internal state to the molecule.*

Show that the detected intensity contains cross terms weighted by the inner product of the internal states carried along the two paths. Explain why unobserved internal excitations can reduce visibility without selecting a definite trajectory. Identify the coherence condition which lets a composite object display the same superposition law as a photon.

2 Symmetry

Exercise 2.1 (Three vibrations of water). **Experiment.** *Infrared spectroscopy finds three fundamental vibrational modes of a water molecule: one bend and two stretches [4]. The equilibrium shape is preserved by a reflection and by interchange of the two hydrogen atoms. Let the resulting finite group G act on the small-displacement space V , and suppose the dynamical operator $D : V \rightarrow V$ commutes with that action.*

Without assigning coordinates to the atoms, show that every eigenspace of D is a G -representation. Decompose V into irreducible representations by averaging an arbitrary

inner product over G . Explain why this decomposition separates the symmetric stretch, anti-symmetric stretch, and bend. Use the transformation of the dipole moment to state which components can absorb infrared radiation and thereby explain the observed selection rule.

Exercise 2.2 (Missing spectral lines). **Experiment.** Atomic spectra contain sharp lines, but many energy differences produce no line. Suppose a symmetry group G acts unitarily on the state space H , the Hamiltonian commutes with G , and the radiation-coupling operator transforms as a representation W .

For initial and final irreducible subspaces $V_i, V_f \subset H$, interpret a transition amplitude as a G -equivariant map

$$V_i \longrightarrow V_f \otimes W.$$

Use Schur's lemma to show that the amplitude vanishes unless V_i occurs in $V_f \otimes W$. Apply this statement to rotational symmetry and an electric dipole, obtaining the usual change in angular momentum. Explain why a vanishing intertwiner predicts an absent line rather than merely a weak one.

Exercise 2.3 (Hydrogen degeneracy). **Experiment.** Ignoring fine structure and external fields, hydrogen energy levels with different orbital angular momentum share the same energy. Spectroscopy organizes the bound-state energy by one principal integer [5]. Let L be angular momentum and A the conserved Runge-Lenz vector.

Starting from their coordinate-free commutator relations on a fixed negative energy eigenspace, rescale A and form two commuting triples

$$J_{\pm} = \frac{1}{2}(L \pm A').$$

Show that each triple generates a copy of $\mathfrak{su}(2)$. Determine the dimension of the resulting representation and recover the degeneracy n^2 . Explain which extra conserved quantity accounts for degeneracy not forced by ordinary spatial rotations, and which experimental perturbations destroy it.

Exercise 2.4 (A full turn changes a sign). **Experiment.** Neutron interferometry detects a relative phase of π when one beam's spin is rotated through 2π , and return after 4π [6]. Spatial directions are transformed by $\text{SO}(3)$, but spin states form rays in a two-dimensional Hilbert space.

Show that a continuous action on rays lifts to an ordinary unitary representation of the double cover $\text{SU}(2)$. Prove that the nontrivial element above the identity rotation acts as $-\text{id}$ in the $\text{spin}-\frac{1}{2}$ representation. Explain why the sign is invisible on an isolated ray yet becomes observable when the rotated beam interferes with an unrotated beam. Identify the obstruction to replacing this projective action by an ordinary representation of $\text{SO}(3)$.

Exercise 2.5 (Protected end states). ***Experiment.** Open integer-spin antiferromagnetic chains exhibit low-energy spin- $\frac{1}{2}$ degrees of freedom at their ends while the bulk remains gapped. Model the ground state by a tensor network whose internal space carries a projective representation u_g of the preserved symmetry group:*

$$u_g u_h = \omega(g, h) u_{gh}.$$

Use associativity to derive the cocycle identity for ω . Show that a change $u_g \mapsto \beta(g)u_g$ changes ω by a coboundary but cannot change its cohomology class. Explain why a nontrivial class permits half-integer edge behavior even though every bulk site has integer spin. Argue why a continuous, symmetry-preserving, gap-preserving deformation cannot remove the edge representation.

Exercise 2.6 (Zeeman splitting). ***Experiment.** A spectral line splits into several components when an atom is placed in a magnetic field, and the pattern depends on the field direction. Let an irreducible rotation representation describe an unperturbed energy eigenspace, and let the field select the subgroup of rotations about its axis.*

Restrict the representation to this subgroup and decompose it into characters. Explain why the perturbation can assign different energies to the resulting weight spaces. Recover the number and spacing of components in the simplest orbital model. Explain why removing the field restores the degeneracy without requiring the eigenvectors to return to a preferred basis.

Exercise 2.7 (Ammonia inversion). ***Experiment.** The ammonia microwave spectrum contains a close doublet associated with the nitrogen atom tunneling between two classically equivalent configurations. Let the reflection exchanging the configurations act unitarily on the low-energy state space and commute with the Hamiltonian.*

Decompose the space into the two characters of the reflection group. Explain why the energy eigenstates are symmetric and antisymmetric combinations rather than states localized in one configuration. Relate the doublet separation to the tunneling amplitude, and explain why a measurement localizing the nitrogen prepares a superposition which subsequently oscillates.

Exercise 2.8 (Parity violation). ***Experiment.** In polarized cobalt-60 decay, electrons are emitted preferentially opposite the nuclear spin. Reflection reverses momentum but not angular momentum.*

Construct the scalar made from the spin and emitted momentum and determine its behavior under spatial reflection. Explain why a reflection-invariant decay law would force the angular distribution to have equal rates in the two directions. Show how the observed asymmetry rules out parity as a symmetry of the weak interaction while leaving rotational covariance intact. Distinguish failure of a physical symmetry from a change of coordinates.

Exercise 2.9 (Identical-particle multiplets). ***Experiment.** Atomic term spectra contain multiplets with systematic absences determined by whether the electrons occupy symmetric or antisymmetric joint states. Let the permutation group act on the tensor power of the one-electron state space.*

Decompose the tensor power into symmetry types. Explain why the fermionic state must lie in the alternating subspace and how this constrains the allowed combination of spatial and spin factors. Derive the singlet–triplet alternative for two electrons. Explain the observed missing terms as absent representation components, not as transitions too weak to detect.

Exercise 2.10 (Raman polarization rules). ***Experiment.** Changing the relative polarizations of incident and scattered light makes some molecular Raman lines appear and others disappear. Let a vibrational mode transform in a representation V , and let the polarizability transform in the symmetric square of the spatial vector representation.*

Express the scattering amplitude as an invariant trilinear form. Determine when such an invariant exists by decomposing the tensor product. Explain the observed polarization-dependent absences as symmetry constraints on the amplitude, and show why rotating the laboratory axes cannot change whether the invariant subspace exists.

3 Fields

Exercise 3.1 (Hertz’s waves). ***Experiment.** Hertz produced and detected electromagnetic waves whose reflection, interference, and propagation speed agreed with Maxwell’s theory [7]. On oriented spacetime, represent the electromagnetic field by a two-form F and current by a three-form J .*

Write Maxwell’s equations using the exterior derivative and the metric’s Hodge operator, without splitting space from time. Show that charge conservation follows from applying the exterior derivative twice. In a source-free contractible region, derive a wave equation for F . Explain which part of the construction needs a metric, which needs only orientation, and why the predicted propagation speed is independent of the observer’s coordinates.

Exercise 3.2 (A charge bends but does not speed up). ***Experiment.** In a uniform magnetic field, a nonrelativistic charged particle follows a circle at constant speed. Let Q be physical space, let F be the magnetic two-form on Q , and let the kinetic energy supply an identification between velocity and momentum.*

Construct the twisted symplectic form

$$\Omega_F = \Omega_{\text{can}} + q \pi^* F$$

on T^*Q . Define the motion by $\iota_{X_H}\Omega_F = dH$ and recover the Lorentz force after choosing Euclidean coordinates only for the final calculation. Show directly from antisymmetry that $dH(X_H) = 0$, explaining the constant speed. Determine what condition on F is forced by $d\Omega_F = 0$, and relate its failure to the hypothetical observation of magnetic charge.

Exercise 3.3 (Phase volume does not compress). **Experiment.** An initially small ensemble of nearly isolated classical systems stretches and folds in phase space but does not change its fine-grained phase volume. Let (M, Ω) be a finite-dimensional symplectic phase space and let φ_t be the flow generated by $H : M \rightarrow \mathbb{R}$.

Use Cartan's formula to prove $\mathcal{L}_{X_H}\Omega = 0$, then prove invariance of $\Omega^n/n!$. Explain how this accounts for the observed incompressibility without choosing canonical coordinates. Contrast the conclusion with a damped oscillator, and identify precisely which structural assumption the damped dynamics violates.

Exercise 3.4 (LIGO's correlated motion). **Experiment.** LIGO observed a time-dependent differential change in the lengths of two perpendicular arms consistent with a passing gravitational wave [8]. Model freely falling mirrors by nearby geodesics with separation field J and tangent u .

Derive the geodesic-deviation equation

$$\nabla_u \nabla_u J + R(J, u)u = 0.$$

Show that the relative acceleration is curvature and hence cannot be removed throughout the apparatus by a change of coordinates. In the weak-field limit, explain why a transverse traceless disturbance lengthens one arm while shortening the other. Identify the measured quantity as a comparison of proper times or optical path lengths, not as the value of a coordinate component.

Exercise 3.5 (The charged weak bosons). **Experiment.** Collider events revealed massive charged $W^\pm$ bosons and a neutral Z^0 boson with the weak-interaction decay patterns predicted by the electroweak model [9, 10]. Let a gauge group G act on a vector bundle $E \rightarrow M$, let A be a connection, and let F_A be its curvature.

Show how a local change of frame transforms A inhomogeneously but transforms F_A by conjugation. Deduce that an invariant pairing on the Lie algebra gives a frame-independent Yang–Mills action. Let a nonzero scalar field select a stabilizer subgroup $H \subset G$. Explain why directions tangent to G/H acquire mass while the unbroken electromagnetic direction remains massless. Relate the resulting charged and neutral directions to the observed particle pattern without treating a gauge choice as an observable.

Exercise 3.6 (Faraday induction). ***Experiment.** A changing magnetic flux through a conducting loop produces an electromotive force around the loop. Let F be the spacetime electromagnetic two-form and let the loop sweep out a cylinder between two times.*

Apply Stokes' theorem to the cylinder and derive Faraday's law as a comparison of fluxes and line integrals. Explain why the result remains meaningful for a moving loop and why no coordinate-dependent electric field components are fundamental to the statement. Determine the sign from the orientation and explain Lenz's law as the experimentally observed energetic response of the circuit.

Exercise 3.7 (Foucault's pendulum). ***Experiment.** The oscillation plane of a long pendulum rotates over a day by an amount which depends on latitude. Treat the slowly transported oscillation direction as a tangent vector on the spherical Earth.*

Parallel transport the direction around the latitude circle and relate its net rotation to the enclosed Gaussian curvature. Recover the observed latitude dependence. Explain why the angle is holonomy rather than a failure of the vector to remain fixed, and separate this geometric effect from local friction and from the choice of longitude coordinates.

Exercise 3.8 (Gravitational redshift). ***Experiment.** Identical clocks placed at different heights in a stationary gravitational field accumulate different proper times, as confirmed by clock-comparison experiments. Let the stationary spacetime possess a timelike symmetry field T .*

Show that the conserved photon energy along a null geodesic is obtained by pairing its tangent with T . Relate the frequencies measured by stationary observers to the norms of their four-velocities and derive the redshift ratio. Explain why the prediction compares proper times and survives all coordinate changes. Recover the leading weak-field height dependence.

Exercise 3.9 (Light is gravitationally lensed). ***Experiment.** Light from a distant source can reach an observer along multiple paths around a massive object, producing several images and measurable time delays. Treat light rays as null geodesics of spacetime.*

Use the variational characterization of null paths to explain the occurrence of stationary optical paths. Show why curvature can produce more than one such path with the same endpoints. Express image distortion through the differential of the endpoint map and identify conjugate points as its singularities. Explain why image positions and relative delays are observables although any drawn coordinate grid is not.

Exercise 3.10 (The Sagnac time difference). ***Experiment.** Two light beams sent in opposite directions around a rotating loop return at different times; ring-laser gyroscopes measure the resulting phase shift. Model the apparatus by a closed world tube and the light beams by its two null directions.*

Derive the proper-time difference measured by the emitter and show that its leading term is proportional to the loop's oriented area and angular velocity. Express the result as a circulation associated with the observer field. Explain why the phase detects rotation of the apparatus and not an arbitrary rotation of spatial coordinates.

4 Topology

Exercise 4.1 (A phase outside the solenoid). **Experiment.** Electron interference fringes shift when magnetic flux through an excluded solenoid changes, even though the magnetic field vanishes along both electron paths [11, 12]. Let X be the field-free region and let A be a closed one-form there.

Show that the relative phase is

$$\exp\left(\frac{iq}{\hbar} \oint_{\gamma} A\right),$$

where γ is the loop formed by the two paths. Prove invariance under $A \mapsto A + df$. Show that the phase depends only on the de Rham class of A and the homology class of γ . Compute it for the complement of an ideal solenoid, and explain why the observed fringe shift detects holonomy rather than a local magnetic force on the electrons.

Exercise 4.2 (A geometric neutron phase). **Experiment.** Polarized neutrons transported through a cyclic sequence of magnetic fields acquire a phase proportional to the solid angle enclosed by the field direction on the sphere [13]. Let $L \rightarrow S^2$ be the line whose fiber is the instantaneous positive-spin eigenspace.

Use orthogonal projection of the trivial connection to construct a connection on L . Calculate its curvature and show that the holonomy around a loop is half the enclosed solid angle, modulo 2π . Explain why changing the phase of each chosen eigenvector changes the local connection form but not the measured interference phase. Determine why no globally smooth nonvanishing choice of positive-spin eigenvector exists.

Exercise 4.3 (Integer Hall plateaus). **Experiment.** At low temperature, a two-dimensional electron system displays Hall conductance locked to integer multiples of e^2/h , with plateaus stable under disorder and small changes of magnetic field [14]. For a gapped periodic model, let the occupied states form a vector bundle E over the Brillouin torus.

Construct the projected connection on E and its curvature F . Show that

$$\frac{i}{2\pi} \int_{\mathbb{T}^2} \text{Tr}(F) \in \mathbb{Z}.$$

Use the response formula to identify this integer with Hall conductance [15]. Prove that it is

unchanged under a continuous deformation that keeps the spectral gap open. Explain both the plateau's stability and the need for a gap closing when the measured integer changes.

Exercise 4.4 (One particle per pump cycle). **Experiment.** An adiabatically driven one-dimensional lattice can transport an integer number of particles per cycle; cold-atom measurements observe this quantized pumping [16]. Treat crystal momentum and drive time as coordinates on a torus, and let E be the occupied-state bundle over that torus.

Relate the transported charge in one period to the curvature of the projected connection on E . Show that the result is its first Chern number and is therefore integral. Explain why small errors in the drive do not change the transported integer while adiabaticity and the energy gap persist. Identify what the experiment would show if the path crossed a degeneracy.

Exercise 4.5 (Zero modes at a boundary). **Experiment.** Topological materials support robust boundary modes whose net direction of propagation is tied to a bulk integer. Let D be the Fredholm operator obtained from the boundary problem, with finite-dimensional kernel and cokernel.

Show that

$$\text{ind } D = \dim \ker D - \dim \text{coker } D$$

is unchanged by sufficiently small perturbations. Relate this analytic integer to the bulk Chern number through an index pairing. Explain why disorder may deform individual boundary states but cannot change the net number of chiral channels without closing the bulk gap or mixing with an opposite boundary. State the experimentally testable transport consequence of this equality.

Exercise 4.6 (Quantized superconducting flux). **Experiment.** Magnetic flux trapped in a superconducting ring occurs in integer multiples of $h/2e$. Let the nonzero condensate define a phase on the ring and let A be the electromagnetic connection.

Show that single-valuedness makes the phase winding an integer. Combine its differential with A to form the gauge-invariant circulation and recover the observed flux unit when the bulk current vanishes. Explain why local changes of condensate phase do not alter the integer. Identify the event required for the ring to move between adjacent flux sectors.

Exercise 4.7 (Weak localization in a ring). **Experiment.** The conductance of a small disordered metallic ring oscillates as magnetic flux through the ring is varied, even when electrons travel diffusively. Pair each closed diffusive path with its time reverse.

Show that the two amplitudes have equal dynamical phase but acquire opposite magnetic holonomies. Derive the flux period of their interference correction. Explain why averaging over disorder suppresses many path pairs but preserves the time-reversed pair. Interpret the oscillation as sensitivity to the connection's holonomy rather than to a force along the wire.

Exercise 4.8 (Graphene’s half-integer offset). ***Experiment.** Graphene exhibits a quantum Hall sequence shifted relative to that of an ordinary parabolic band. Near a band-touching point, the low-energy eigenspaces form lines over a loop in momentum space.*

Transport an eigenline around the loop and compute its geometric phase. Explain how the resulting sign changes the semiclassical quantization rule and produces the observed Hall offset. Show that the phase cannot be removed by choosing eigenvectors smoothly along the entire loop while respecting single-valuedness. State what perturbation can gap the touching point and alter the inference.

Exercise 4.9 (A topological interface mode). ***Experiment.** Engineered one-dimensional lattices with alternating couplings display a localized mode where the coupling pattern switches order. Let the gapped bulk Hamiltonian possess a symmetry which pairs positive and negative energies.*

Associate a winding integer to each translation-invariant bulk. Show that the two coupling orders have different integers. Explain why an interface must then support a midgap state and why its wave function is localized. Determine which perturbations can move or remove the state, and relate the answer to whether the protecting symmetry and bulk gap survive.

Exercise 4.10 (Josephson voltage oscillations). ***Experiment.** A constant voltage across a Josephson junction produces an alternating supercurrent whose frequency obeys $f = 2eV/h$. Let the two superconductors carry condensate phases connected by the electromagnetic gauge field.*

Construct the gauge-invariant phase difference across the junction. Use its time evolution to derive the measured frequency–voltage relation. Explain why the coefficient reveals charge $2e$, why a local phase convention cannot alter the prediction, and how phase winding makes the oscillation a global comparison rather than an observable absolute phase.

5 Noncommutativity

Exercise 5.1 (An entangled state has no local state vector). ***Experiment.** Two-qubit tomography reconstructs joint states whose correlations cannot be represented by assigning a pure state to each qubit. Let ρ be the reconstructed positive trace-one operator on $H_A \otimes H_B$. Derive the unique operator ρ_A satisfying*

$$\text{Tr}(\rho_A X) = \text{Tr}(\rho(X \otimes \text{id}))$$

for every observable X on H_A . For a maximally entangled pure state, show that ρ_A is maximally mixed. Explain how a globally sharp state can therefore give locally random results.

Show that no choice of bases in H_A and H_B turns this state into a product, and identify the basis-free invariant that detects this failure.

Exercise 5.2 (An ancilla improves discrimination). **Experiment.** Entangling a probe with an untouched ancilla can increase the probability of distinguishing two laboratory channels. For channels Φ_0, Φ_1 , the largest unassisted bias is governed by their action on system states, while the assisted bias allows states on $H \otimes A$.

Express the assisted bias as the norm of $(\Phi_0 - \Phi_1) \otimes \text{id}_A$. Show that an ancilla of dimension $\dim H$ is sufficient. Construct a pair of channels for which the assisted value is strictly larger than the unassisted value. Explain why the experiment forces a matrix-level norm on maps rather than a norm which tests only the original system.

Exercise 5.3 (Bell correlations). **Experiment.** Loophole-reduced Bell tests observe correlations violating the CHSH bound 2 while remaining below the quantum bound $2\sqrt{2}$ [17]. Let A_0, A_1 and B_0, B_1 be self-adjoint contractions belonging to commuting local observable algebras, and set

$$C = A_0(B_0 + B_1) + A_1(B_0 - B_1).$$

Prove the classical bound when all four observables commute. In the quantum case, compute C^2 and derive $\|C\| \leq 2\sqrt{2}$ [18]. Give observables and a bipartite state attaining equality. Explain which algebraic change permits the experimentally observed violation and why the same calculation still forbids correlations beyond the quantum bound.

Exercise 5.4 (Two photons leave together). **Experiment.** When two indistinguishable photons enter opposite ports of a balanced beam splitter simultaneously, coincident detections at the two outputs are suppressed [19]. Let H be the one-photon mode space and describe two identical bosons in the symmetric square of H .

Apply the beam-splitter unitary functorially to the two-photon state. Show that the two amplitudes leading to one photon in each output cancel, whereas the amplitudes for both photons in one output add. Explain why distinguishable photons do not show complete suppression. Relate the width and depth of the observed coincidence dip to temporal overlap and indistinguishability rather than to a choice of mode coordinates.

Exercise 5.5 (Sequential spin measurements). **Experiment.** A beam selected by a vertical Stern–Gerlach analyzer becomes uncertain when measured horizontally; selecting the horizontal output then restores two possible vertical outcomes. Represent the selections by rank-one projections P and Q .

Compute the sequential probabilities from products such as PQP . Show that the observed order effect disappears exactly when the projections commute. Explain why the disturbance is not

accounted for by ignorance of a pre-existing pair of spin components. Express the uncertainty through the commutator without selecting matrices until the final numerical example.

Exercise 5.6 (No perfect photocopier). **Experiment.** Optical experiments can approximately clone states from a known ensemble, but no device produces two perfect copies of every unknown polarization state. Suppose a physical evolution U and a fixed blank state b satisfied

$$U(v \otimes b) = v \otimes v$$

for every unit vector v .

Compare inner products before and after the evolution and derive a contradiction for two distinct nonorthogonal states. Explain why known orthogonal states can be copied and why probabilistic or approximate cloning does not contradict the argument. Identify preservation of transition probabilities as the structure forced by the experiment.

Exercise 5.7 (Teleportation correlations). **Experiment.** Quantum teleportation transfers an unknown qubit state using a shared entangled pair, two classical bits, and no transmission of the original carrier. Let H be the qubit space and use the canonical correspondence between maps on H and vectors in $H \otimes H^*$.

Decompose the combined state into four orthogonal measurement alternatives. Show that each alternative leaves the remote system related to the input by a known unitary correction. Explain why averaging before receipt of the classical message gives a maximally mixed remote state, preventing faster-than-light signaling. State what experimental statistics certify fidelity beyond a classical measure-and-prepare scheme.

Exercise 5.8 (Decoherence of interference). **Experiment.** Interference visibility falls when path information leaks into an environment, even if no observer reads that information. Let the two paths become correlated with normalized environment states e_0, e_1 .

Take the partial trace of the joint pure state and show that the off-diagonal interference term is multiplied by $\langle e_1, e_0 \rangle$. Relate fringe visibility to this overlap and path distinguishability to its complement. Explain the disappearance of fringes when the records are orthogonal and their recovery under a quantum eraser which recombines the record alternatives.

Exercise 5.9 (Quantum Zeno inhibition). **Experiment.** Repeated measurements of whether an unstable or coherently driven system remains in its initial state can inhibit its transition. Let P project onto that state and let $U(t)$ be the unobserved evolution.

Show that after N equally spaced ideal tests the survival probability is

$$\| (PU(t/N)P)^N v \|^2.$$

Use the short-time quadratic behavior to find its limit as $N \rightarrow \infty$. Explain why the result depends on the noncommuting alternation of evolution and projection, and state which finite measurement times limit the laboratory effect.

6 Braiding

Exercise 6.1 (Fractional shot noise). ***Experiment.** Shot-noise measurements in a fractional quantum Hall state infer mobile charges of magnitude $e/3$, smaller than the electron charge [20]. Suppose rare tunneling events transfer quasiparticle charge q independently across a constriction.*

Derive the low-frequency relation between current noise and mean tunneling current for a Poisson process. Show how the measured ratio determines $|q|$ without assigning a microscopic path to each event. Explain why the result supports fractionalized excitations rather than electrons arriving in a regular correlated train, and state the additional assumptions needed for that inference.

Exercise 6.2 (An anyonic exchange phase). ***Experiment.** Electronic interferometry in the $\nu = 1/3$ fractional quantum Hall regime shows phase slips consistent with the statistical phase of Abelian anyons [21]. The configuration space of indistinguishable particles in a plane has braid group B_n as its fundamental group.*

Show that a one-dimensional unitary representation of B_n assigns the same phase $e^{i\theta}$ to every elementary exchange. Compare this with the only one-dimensional exchange possibilities for particles in three spatial dimensions. Compute the change in interference phase when one quasiparticle winds around another, carefully distinguishing exchange from a full winding. Explain how discrete phase slips can reveal θ even when ordinary electromagnetic flux also contributes to the total phase.

Exercise 6.3 (Fractional collision statistics). ***Experiment.** Two dilute beams of $e/3$ quasiparticles sent into an electronic beam splitter produce current correlations intermediate between bosonic bunching and fermionic antibunching [22]. Model the incoming particles by Abelian exchange phase $e^{i\theta}$.*

List the indistinguishable two-particle alternatives contributing to an output coincidence. Show where their relative exchange phase enters the current cross-correlation. Explain why the measured dependence on source current and splitter transmission can determine θ , and recover $\theta = \pi/3$ for the reported data. State which interaction and independence assumptions are required before calling the correlation a statistics measurement.

Exercise 6.4 (Mach–Zehnder braiding). ***Experiment.** A chiral electronic Mach–Zehnder*

interferometer in the fractional quantum Hall regime displays a phase dependence consistent with braiding an interfering $e/3$ quasiparticle around localized quasiparticles [23]. Represent the two paths by amplitudes and the winding by a unitary character of the braid group.

Separate the electromagnetic and statistical contributions to the measured phase. Explain why adding one enclosed quasiparticle shifts the interference pattern by the double-exchange phase. Derive the predicted shift for $\theta = \pi/3$. Explain how the two-path geometry and independent charge measurement help rule out a purely Coulombic explanation.

Exercise 6.5 (Fermionic antibunching). **Experiment.** Two identical fermions incident on a two-input beam splitter show suppressed coincident occupation of the same output mode, opposite to bosonic bunching. Represent their joint state in the exterior square of the one-particle mode space.

Apply the one-particle scattering unitary to the exterior product and compute the output probabilities. Explain why equal one-particle states wedge to zero. Show how partial distinguishability weakens the suppression, and identify exchange antisymmetry rather than a repulsive force as the explanation of the ideal effect.

Exercise 6.6 (Quasiparticle charge from flux insertion). **Experiment.** Adiabatically increasing flux through a puncture in a fractional Hall fluid transports charge between its boundaries. Regard the family of gapped ground states as a bundle over the flux parameter.

Relate the transported charge to the Hall response and hence to the experimentally measured filling fraction. Explain why one flux quantum creates a localized excitation carrying fractional charge. Show why the argument depends on the bulk gap and gauge equivalence after a full flux cycle, and connect this transported charge with the shot-noise measurement.

Exercise 6.7 (Coherent anyon bunching). **Experiment.** An electronic interference experiment observes coherent bunching and dissociation of fractional quantum Hall anyons [24]. Model the paths by amplitudes whose relative phase includes both propagation and fractional exchange.

Identify the alternatives which end in the same detected charge configuration and must therefore be added before squaring. Show how the exchange phase changes enhancement into suppression as the control phase is varied. Explain why simultaneous measurements of currents and correlations separate coherent anyon processes from independent Poisson events. State which loss of path coherence would erase the effect.

Exercise 6.8 (Stable individual anyons). **Experiment.** A graphene Fabry–Perot interferometer resolves configurations containing a fixed number of localized anyons and finds them stable for minutes or hours while the interference phase records their statistics [25]. Let the localized configuration label a sector of the interferometer’s state space.

Explain why a transition between sectors produces a discrete phase jump rather than a continuous drift. Relate the jump to the holonomy of an edge anyon winding around the localized sector. Show how long dwell times allow separate estimates of within-sector interference and sector-changing rates. Explain what the stability demonstrates and why it does not by itself establish non-Abelian fusion.

Exercise 6.9 (Thermal transport at a Hall edge). **Experiment.** *Heat carried along a fractional quantum Hall boundary is observed to be approximately quantized in units set by temperature and fundamental constants, with the sign recording net propagation direction [26]. Associate a chiral contribution to each protected edge sector.*

Show why independent sectors add while oppositely directed sectors subtract. Relate the net coefficient to the boundary theory's chiral invariant. Explain how the measurement constrains candidate quasiparticle theories even when they have the same electrical Hall conductance. Identify equilibration and heat leakage as assumptions needed before interpreting the measured value topologically.

References

- [1] T. Young, *The Bakerian Lecture: Experiments and calculations relative to physical optics*, Philosophical Transactions of the Royal Society of London **94** (1804), 1–16.
- [2] W. Gerlach and O. Stern, *Der experimentelle Nachweis der Richtungsquantelung im Magnetfeld*, Zeitschrift für Physik **9** (1922), 349–352.
- [3] J. B. Altepeter et al., *Ancilla-assisted quantum process tomography*, Physical Review Letters **90** (2003), 193601.
- [4] T. Shimanouchi, *Tables of Molecular Vibrational Frequencies, Consolidated Volume I*, National Bureau of Standards, 1972.
- [5] A. Kramida, *A critical compilation of experimental data on spectral lines and energy levels of hydrogen, deuterium, and tritium*, Atomic Data and Nuclear Data Tables **96** (2010), 586–644.
- [6] R. Hagiwara et al., *Phase shift of a neutron wave function by 4π spin rotation*, Physical Review Letters **65** (1990), 3181–3184.
- [7] H. Hertz, *Über sehr schnelle elektrische Schwingungen*, Annalen der Physik **267** (1887), 421–448.

- [8] B. P. Abbott et al., *Observation of gravitational waves from a binary black hole merger*, Physical Review Letters **116** (2016), 061102.
- [9] G. Arnison et al., *Experimental observation of isolated large transverse energy electrons with associated missing energy at $\sqrt{s} = 540$ GeV*, Physics Letters B **122** (1983), 103–116.
- [10] G. Arnison et al., *Experimental observation of lepton pairs of invariant mass around 95 GeV/c² at the CERN SPS collider*, Physics Letters B **126** (1983), 398–410.
- [11] Y. Aharonov and D. Bohm, *Significance of electromagnetic potentials in the quantum theory*, Physical Review **115** (1959), 485–491.
- [12] A. Tonomura et al., *Evidence for Aharonov–Bohm effect with magnetic field completely shielded from electron wave*, Physical Review Letters **56** (1986), 792–795.
- [13] T. Bitter and D. Dubbers, *Manifestation of Berry’s topological phase in neutron spin rotation*, Physical Review Letters **59** (1987), 251–254.
- [14] K. von Klitzing, G. Dorda, and M. Pepper, *New method for high-accuracy determination of the fine-structure constant based on quantized Hall resistance*, Physical Review Letters **45** (1980), 494–497.
- [15] D. J. Thouless, M. Kohmoto, M. P. Nightingale, and M. den Nijs, *Quantized Hall conductance in a two-dimensional periodic potential*, Physical Review Letters **49** (1982), 405–408.
- [16] M. Lohse et al., *A Thouless quantum pump with ultracold bosonic atoms in an optical superlattice*, Nature Physics **12** (2016), 350–354.
- [17] B. Hensen et al., *Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres*, Nature **526** (2015), 682–686.
- [18] B. S. Tsirelson, *Quantum generalizations of Bell’s inequality*, Letters in Mathematical Physics **4** (1980), 93–100.
- [19] C. K. Hong, Z. Y. Ou, and L. Mandel, *Measurement of subpicosecond time intervals between two photons by interference*, Physical Review Letters **59** (1987), 2044–2046.
- [20] R. de-Picciotto et al., *Direct observation of a fractional charge*, Nature **389** (1997), 162–164.
- [21] J. Nakamura et al., *Direct observation of anyonic braiding statistics*, Nature Physics **16** (2020), 931–936.

- [22] H. Bartolomei et al., *Fractional statistics in anyon collisions*, Science **368** (2020), 173–177.
- [23] H. K. Kundu et al., *Anyonic interference and braiding phase in a Mach–Zehnder interferometer*, Nature Physics **19** (2023), 515–521.
- [24] B. Ghosh et al., *Coherent bunching of anyons and dissociation in an interference experiment*, Nature **642** (2025), 922–927.
- [25] N. L. Samuelson et al., *Slow quasiparticle dynamics and anyonic statistics in a fractional quantum Hall Fabry–Pérot interferometer*, Physical Review X **16** (2026), 011062.
- [26] M. Banerjee et al., *Observed quantization of anyonic heat flow*, Nature **545** (2017), 75–79.
- [27] E. Witten, *Quantum field theory and the Jones polynomial*, Communications in Mathematical Physics **121** (1989), 351–399.
- [28] N. Reshetikhin and V. Turaev, *Invariants of 3-manifolds via link polynomials and quantum groups*, Inventiones Mathematicae **103** (1991), 547–597.